Exposure Is Not Adoption

What 129 months of U.S. labour-market data can and cannot say about generative AI

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Abstract

Occupational exposure indices now routinely appear in policy writing as if they measured labour-market outcomes. They do not. An exposure score measures whether a language model can perform tasks that an occupation contains; it says nothing about whether any employer has adopted the model, at what price, for which tasks, or with what effect on headcount. This paper separates the two questions using 129 months of U.S. Bureau of Labor Statistics data, January 2016 to September 2026, covering the first four years of widely available generative AI.

Three findings stand out. First, the aggregate labour market shows no displacement regime: unemployment is 4.2 per cent, payroll growth is 0.31 per cent year over year, and job openings per unemployed person have fallen from a March 2022 peak of 2.04 to 1.01 — a cooling, not a collapse. Second, one sector does break sharply. Information-sector employment peaked in November 2022, the month ChatGPT was released, at 108.8 on a 2019 = 100 index, and has fallen to 95.6 — 12.1 per cent below the peak and 0.8 per cent below its January 2016 level. Extending the pre-October 2022 trend produces a counterfactual of 111.6 for September 2026, a gap of 15.9 index points, or 14.3 per cent. Third, real wages have not moved the way a displacement story predicts: average real wage growth after November 2022 is 0.59 per cent per year, against 0.82 per cent in 2016–2019, and the latest reading is −0.69 per cent.

The information-sector break is consistent with the displacement thesis and equally consistent with the interest-rate cycle and with the unwinding of pandemic-era over-hiring, all of which begin in the same quarter. Sector aggregates cannot separate them. What would separate them is not more exposure scoring but adoption data at the establishment and task level, hiring composition by seniority, and within-occupation wage dispersion. The paper states five falsifiable predictions with explicit decision rules, so that the next four years of data can adjudicate the question that the exposure literature has so far only posed.

Keywords: generative AI, labour demand, occupational exposure, task-based models, measurement, identification

JEL: J21, J23, J24, O33

1. Introduction

1.1 The claim under test

The public argument about generative AI and work runs through a small number of numbers. Roughly four in five U.S. workers hold an occupation in which at least a tenth of tasks could be performed by a language model; roughly one in five holds one in which half of tasks could be. About 40 per cent of global employment is exposed on one widely reported estimate, and 60 per cent in advanced economies. These are exposure estimates. They answer a question about capability: given the tasks an occupation contains, how much of that task bundle lies inside the current capability of a model?

The policy conversation then borrows them to answer a different question: what will happen to employment, wages, and the distribution of income? That step is not licensed by the measurement. Between a task that a model can perform and a job that an employer eliminates sit adoption cost, integration cost, liability, quality risk, organisational inertia, regulation, and a price for the model relative to the wage it would replace. None of those appear in an exposure index.

This paper takes the gap seriously and asks what the data can actually show. I use a single, long, public data set — 129 monthly observations of U.S. sector employment, job openings, unemployment, and wages — to ask three questions:

1. Is there a labour-market break in the generative-AI period that is large enough and sharp enough to be visible at sector level?
2. If there is, can it be attributed to AI rather than to the other shocks that begin at the same moment?
3. If it cannot, what evidence would attribute it, and what should a reader do with the exposure numbers in the meantime?

The answers are: yes, in one sector; no, not from this data; and here is a testable agenda.

1.2 Why the timing is unusually clean and unusually dirty

The generative-AI period has an unusually sharp public starting point. ChatGPT was released on 30 November 2022; the November 2022 employment report, published in early December, covers a payroll period that ends before the release. The first full month of potential labour-market effect is therefore December 2022 or January 2023. This is rare in technology economics, where adoption is gradual and the treatment date is contested.

It is also unusually dirty, for three reasons that all point the same way.

First, the interest-rate cycle. The Federal Reserve began raising rates in March 2022 and continued through July 2023. Technology firms, which are the most information-sector-intensive employers, cut hiring and headcount in response to the cost of capital, not only to AI. The information sector's employment peak in November 2022 is therefore coincident with both a technology shock and a financial shock.

Second, the unwinding of pandemic over-hiring. The information sector index rose from 96.4 in January 2016 to 108.8 in November 2022, a 12.8 per cent increase, much of it concentrated in 2020–2022 as digital demand surged. Some of the subsequent decline is a return to trend rather than a new shock.

Third, composition. The information sector is small — about 2 per cent of nonfarm payrolls — and its employment is dominated by a handful of large firms whose hiring decisions are visible, discretionary, and responsive to equity prices. A sector that represents 2 per cent of employment can show a large percentage decline with almost no effect on the aggregate.

All three point in the same direction: the information-sector break is real and large, and it is not identified as an AI effect by the data that reveal it.

1.3 What this paper does not do

It does not estimate a causal effect of AI on employment. The data cannot support that estimate, and a paper that reported one would be reporting a modelling assumption. It does not build a new exposure index; several exist, they disagree, and building another would not resolve the adoption question. It does not forecast. The counterfactual in Section 5 is an accounting device for sizing the break, not a prediction of what would have happened.

1.4 Contributions

1. A quantitative description of the post-November-2022 break in U.S. sector employment, with the size of the break measured against a pre-trend counterfactual and against sector controls.
2. A demonstration that the three candidate explanations — AI adoption, the rate cycle, pandemic normalisation — are observationally equivalent in sector aggregates, and an explicit statement of what would break the tie.
3. Five falsifiable predictions with numerical decision rules, so that the argument can be settled by data rather than by priors.
4. A reproducible analysis: one CSV, one script, no proprietary data, every number in this paper generated by the script in the repository.

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2. What exposure indices measure

2.1 Task overlap is a capability measure

The modern exposure literature descends from the task-based framework of Acemoglu and Restrepo (2019). Technology displaces labour from some tasks and reinstates it in others; the net employment effect depends on the relative strength of the two, and on the demand elasticity of the affected tasks. The framework is about tasks, not occupations, and it is explicitly an equilibrium framework: displacement in one task can raise demand for labour in another.

Eloundou et al. (2023) operationalise the framework for language models. They map occupations to tasks, ask whether a model can perform each task, and aggregate to an occupation-level share. Their headline estimates are that about 80 per cent of the U.S. workforce could have at least 10 per cent of tasks affected, and about 19 per cent could have at least 50 per cent affected. The paper is careful about what this means: it is a measure of *potential* impact given current model capability, explicitly not a prediction of job loss. The distinction is stated in the paper and is routinely dropped in citation.

Tolan et al. (2021) build a related but distinct measure, mapping tasks to cognitive abilities and then to AI benchmark performance. The unit is the task and the ability, not the occupation, and the resulting index answers a narrower question: how close is benchmark performance to the ability requirements of this task? Zarifhonarvar (2023) applies an LLM-specific exposure measure to occupations and reports a similar distribution.

These measures differ in construction, in the capability frontier they assume, and in whether they weight tasks by time spent. They share one property: none of them observes an employer choosing not to hire.

2.2 The units-of-analysis problem

Table 1 lists the eight sources this paper works from, with the unit of analysis of each and the claim each can support. The table is the argument. A paper whose unit is the occupation cannot support a claim about firms; a paper whose unit is the firm cannot support a claim about occupations; a paper whose unit is the task can support neither without an aggregation assumption.

Table 1. Units of analysis in the AI-and-labour literature

Source	Unit of analysis	What it measures	What it cannot show
Acemoglu & Restrepo (2019)	Task	Displacement and reinstatement forces	Magnitudes, timing
Eloundou et al. (2023)	Occupation, task share	Model capability against task content	Adoption, employer choice
Acemoglu et al. (2022)	Establishment, vacancy	Hiring behaviour of AI-exposed establishments	Occupations below establishment level
Acemoglu (2024)	Economy, task-based macro model	Bounded aggregate TFP effect	Distribution, transition costs
Cazzaniga et al. (2024)	Country, occupation	Cross-country exposure	Within-country incidence
Tolan et al. (2021)	Task, cognitive ability	Capability-to-requirement distance	Employer behaviour
Eisfeldt et al. (2023)	Firm, equity valuation	Market-implied exposure	Employment
Zarifhonarvar (2023)	Occupation	LLM-specific exposure	Adoption, wages

Two of the eight observe behaviour rather than capability. Acemoglu et al. (2022) use establishment-level vacancy postings, which is the closest thing in the set to an adoption measure: a vacancy is a firm spending money to hire. Eisfeldt et al. (2023) use equity valuations, which are forward-looking and therefore contain beliefs about adoption rather than adoption itself. The other six measure capability, exposure, or a model of the economy.

2.3 Where the literature actually disagrees

The disagreement is not about whether language models can perform many tasks. It is about the size of the aggregate effect and the speed of adjustment.

Acemoglu (2024) builds a task-based macro model and bounds the total factor productivity effect of AI within ten years at well under one per cent, with a correspondingly small GDP effect, on the argument that the tasks that are both exposed and economically significant are a minority of the task bundle. This is the strongest available statement of the modest-effects position, and it is a model result: its conclusion depends on the share of tasks assumed automatable and on the cost savings assumed.

The IMF's 2024 staff discussion note takes a different cut: exposure is large (about 40 per cent of global employment, 60 per cent in advanced economies), and roughly half of exposed employment could be complemented rather than substituted, with distributional effects that vary by income level. This is a statement about exposure and its composition, not about realised employment.

Acemoglu et al. (2022) provide the closest thing to behavioural evidence, and their period predates generative AI: establishment-level vacancy data through 2018, in which AI-exposed establishments increase AI-related hiring without a visible reduction in other hiring. If that finding holds forward, it supports the complementarity

channel; if it does not, the burden shifts to explaining why.

The honest summary is that the literature contains a well-specified framework, several capability measures, one behavioural measurement at establishment level, and one macro model with a small bound. None of it observes the adoption decision for generative AI at the task level. That is the gap this paper's data can narrow but not close.

3. Data

3.1 Sources

All data are from the U.S. Bureau of Labor Statistics, retrieved through the public API. The series are listed in Table 2.

Table 2. BLS series

Series ID	Definition	Use
LNS14000000	Unemployment rate, per cent, seasonally adjusted	Aggregate slack
LNS13000000	Unemployed level, thousands, SA	Denominator of V/U
JTS000000000000000JOL	Job openings, thousands, SA	Numerator of V/U
CES0000000001	Total nonfarm payrolls, thousands	Aggregate employment
CES0500000003	Average hourly earnings, total private	Wages
CUUR0000SA0	CPI-U, all items, NSA	Real wages
CES3000000001	Manufacturing employment	Sector
CES5000000001	Information employment	Sector
CES6000000001	Professional and business services employment	Sector
CES6500000001	Health care and social assistance employment	Sector

3.2 Transformations

Sector employment is expressed as an index with the 2019 calendar-year average equal to 100. The 2019 base is deliberate: it is the last full pre-pandemic year and it predates both the generative-AI period and the post-pandemic labour-market tightness, so the index does not conflate the pandemic shock with the later one. Twelve-month changes are computed from the index. Real wage growth is

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g_real = (1 + g_nominal) / (1 + cpi) − 1
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where g_nominal is the twelve-month change in average hourly earnings and cpi is the twelve-month change in CPI-U. Openings per unemployed person is the ratio of openings to unemployed, with both series seasonally adjusted. JOLTS is published with a one-month lag, so the most recent month in the file carries no openings observation; the last valid V/U reading is August 2026.

Two data notes matter for interpretation. CPI-U is not seasonally adjusted while average hourly earnings are; twelve-month changes are essentially unaffected by the mismatch, and the paper uses twelve-month changes throughout. BLS revises payroll and wage series twice after first publication; the analysis uses the current vintage, which is the correct choice for a retrospective but means the most recent months are provisional.

3.3 Coverage

The panel covers January 2016 to September 2026: 129 monthly observations, spanning the pre-pandemic expansion, the pandemic contraction and recovery, the 2021–2022 labour-market tightening, and the generative-AI period. The treatment month used throughout is November 2022, the month of ChatGPT's release. Because the payroll survey's reference period ends before the release, any true effect must appear from December 2022 onward; the November date is therefore conservative — it attributes to AI a month in which AI cannot have had an effect, which if anything biases the measured break upward.

4. Method

4.1 Descriptive estimates

For each sector I report the index level, the date and value of the peak, the percentage change from the peak to the latest observation, and the twelve-month growth rate. These are descriptive and require no identifying assumption beyond the accuracy of the underlying series.

4.2 The pre-trend counterfactual

To size the break, I fit a linear trend to each sector index over January 2016 to October 2022 and extend it to September 2026. The counterfactual answers: if the sector had continued at its pre-November-2022 average monthly growth rate, where would it be? The gap between the counterfactual and the actual is the part of the movement not explained by the pre-existing trend.

Three caveats are load-bearing. A linear trend is a crude summary; sector employment is not linear over a decade, and the 2016–2022 period includes the pandemic, whose recovery is not a trend. Extending a trend for 47 months compounds any error in the slope. And the counterfactual is not a forecast of what would have happened absent AI: absent AI, rates would still have risen, and the sector would still have corrected. The gap is therefore an upper bound on any AI-attributable component, not an estimate of it.

4.3 Difference-in-differences

As a second cut I compute, for each sector, the log change in the index from October 2022 to September 2026, and subtract the same quantity for health care and social assistance. Health care is the natural control in this data because it is large, domestically driven, exposed to AI in the exposure indices (clinical documentation, claims, scheduling), and yet structurally supply-constrained: an ageing population and a staffing shortage mean that employment grows almost regardless of technology.

The control is imperfect in a way that biases the estimate. Health care's growth is driven by demographics that have nothing to do with the counterfactual for information-sector employment; subtracting it therefore attributes to treatment both the treatment effect and the difference between two unrelated secular trends. The number is reported because it is informative about magnitude, not because it identifies a causal parameter.

4.4 What would count as evidence

For the displacement thesis — exposure reduces employment in exposed occupations — the sector data would need to show a break that is (a) coincident with adoption rather than with the rate cycle, (b) concentrated in task-exposed occupations within sectors rather than across sectors, and (c) visible in hiring flows before it is visible in headcount, because firms stop hiring before they fire. None of these can be tested with sector aggregates. Section 7 states what data would test them.

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5. Results

5.1 The aggregate labour market shows no displacement regime

As of September 2026 the unemployment rate is 4.2 per cent, up from a low of 3.4 per cent in April 2023 but below its 2016–2019 average. Total nonfarm payroll growth is 0.31 per cent year over year, and the post-November-2022 average is 1.16 per cent — positive throughout. The openings-to-unemployed ratio is 1.01 as of August 2026, down from 2.04 in March 2022 and roughly back to its pre-pandemic average of 1.07.

Figure 5 shows the path. The labour market tightened violently through 2021 and 2022, peaked in the first quarter of 2022 — three quarters before ChatGPT — and has loosened steadily since. A labour market in which openings exceed unemployed persons by 100 per cent at the peak and by 1 per cent at the trough is a labour market returning to balance, not one being displaced by a technology.

If generative AI were producing aggregate displacement of the scale implied by the 19-per-cent-highly-exposed estimate, the first place it should appear is in the separation and hiring rates of exposed occupations, and the second is in the aggregate unemployment rate. The aggregate shows neither. That is weak evidence against displacement at scale, because four years is short and adoption is uneven; it is not evidence that displacement is absent in specific occupations.

5.2 The information sector breaks

Information is the one sector with a break that is both large and sharply timed. The index rose from 96.4 in January 2016 to a peak of 108.8 in November 2022 — the treatment month — then fell to 95.6 in September 2026. That is 12.1 per cent below the peak and 0.8 per cent below where the sector started in 2016: fifteen years of employment growth erased in 47 months.

Figure 1 shows the four sector indices with the treatment month marked. The information line is the only one that turns down at the marker.

Figure 2 adds the pre-trend counterfactual. The pre-November-2022 trend is +0.86 index points per year. Extended 47 months, it predicts 111.6 for September 2026. The actual is 95.6. The gap is 15.9 index points, or 14.3 per cent of the counterfactual.

Figure 3 shows twelve-month growth rates. The information series first turns negative in May 2023, six months after the treatment month, and reaches −4.2 per cent in September 2026. But the growth rate had already peaked at 9.96 per cent in August 2021 and was falling steadily through 2022 — a fact Section 6.1 takes up, because it changes what the level peak means. Its trough in the sample is −8.8 per cent in June 2020, during the pandemic shutdown — so the current decline is roughly half as deep as the pandemic shock, and unlike the pandemic shock it is persistent.

5.3 The other sectors do not

Professional and business services peaked later, in May 2023, at 106.9, and has drifted to 105.2 — 1.6 per cent below peak, still 12.2 per cent above 2016, and growing 0.55 per cent year over year. Manufacturing peaked in January 2023 at 101.0 and sits at 99.0, 1.9 per cent below peak and growing 0.32 per cent. Health care has not peaked: it rose from 92.5 in January 2016 to 115.9 in September 2026, 25.3 per cent higher, and is still growing 1.88 per cent year over year.

The pattern matters for interpretation. If a general-purpose technology were displacing labour economy-wide, the sector with the highest measured exposure — professional and business services, which contains legal, accounting, consulting, and administrative support work — should show the largest break. It shows a smaller one than information. The sector that shows the break is the one that produces the technology, not the one whose tasks the technology performs. That is more consistent with a capital-market and over-hiring story than with a task-displacement story, though it does not prove either.

5.4 Wages: the surprise

The displacement story predicts downward pressure on wages in exposed occupations. Aggregate real wage growth shows no such pressure at the scale of the economy, and its direction is the opposite of what a labour-demand collapse would produce.

Average real wage growth was 0.82 per cent per year in 2016–2019, 1.61 per cent in 2020–2021, and 0.59 per cent after November 2022. Nominal growth post-treatment averages 4.04 per cent against CPI inflation of 3.44 per cent — a thin real margin. Twelve of the 46 post-treatment months show negative real growth, and the latest reading is −0.69 per cent.

Figure 4 plots nominal and real growth. The striking feature is not the level but the volatility: real wage growth swings between roughly −3 and +3 per cent through 2022–2024 as inflation moves, and the recent negative reading is a product of nominal wage growth settling near 4 per cent while CPI growth remains above 3 per cent. This is a wage-price adjustment story. It is not a story about AI eroding the bargaining power of exposed workers, which would show as a widening gap between exposed and unexposed occupations — a comparison the sector data cannot make at all.

5.5 Summary of the break

Table 3. Sector results, January 2016 to September 2026

Sector	Peak (date, index)	Latest (index)	From peak	Since Jan 2016	12-month	Pre-trend (pts/yr)	Post-trend (pts/yr)	Counterfactual gap
Information	Nov 2022, 108.8	95.6	−12.1%	−0.8%	−4.20%	+0.86	−2.84	−15.9 pts (−14.3%)

Sector	Peak (date, index)	Latest (index)	From peak	Since Jan 2016	12-month	Pre-trend (pts/yr)	Post-trend (pts/yr)	Counterfactual gap
Professional & business	May 2023, 106.9	105.2	−1.6%	+12.2%	+0.55%	+1.31	−0.57	−6.5 pts (−5.8%)
Manufacturing	Jan 2023, 101.0	99.0	−1.9%	+2.4%	+0.32%	+0.13	−0.77	−2.3 pts (−2.3%)
Health care	Sep 2026, 115.9	115.9	0.0%	+25.3%	+1.88%	+0.81	+3.69	+10.8 pts (+10.3%)

Table 4. Difference-in-differences against health care, log change October 2022 to September 2026

Sector	Log change	Health care log change	Difference
Information	−0.124	+0.128	**−25.2 pp**
Professional & business	−0.013	+0.128	−14.1 pp
Manufacturing	−0.018	+0.128	−14.6 pp

The difference-in-differences numbers are large and should be read with the caution in Section 4.3: health care is growing for demographic reasons, so a substantial part of every difference is the control's secular growth, not the sector's decline. The information estimate is the one with a plausible mechanism; the professional and manufacturing estimates are better read as the size of the health-care trend than as evidence of AI effects.

6. The identification problem

6.1 The turn predates the treatment

The most important number in this paper is not the size of the information-sector decline. It is the date on which the decline began, measured as a rate of change rather than a level.

Twelve-month growth in information-sector employment peaked at 9.96 per cent in August 2021. It was 9.46 per cent in October 2021, 8.63 per cent in January 2022, 8.51 per cent in April 2022, 7.39 per cent in July 2022, and 5.51 per cent in October 2022. In November 2022 — the treatment month — it was 5.34 per cent and falling. It crossed zero in May 2023.

In other words, the information sector's hiring rate began decelerating fifteen months before ChatGPT was released, and had lost more than four percentage points of annual growth before the release. The level series peaks in November 2022 because a decelerating series that has not yet crossed zero still rises to a peak. The peak date is an artefact of the level series; the turn is a fact about the growth series, and the growth series turns in the third quarter of 2021.

What was happening in the third quarter of 2021? Inflation had begun to run above target; the Federal Reserve was signalling that tapering would begin; and the technology sector was working through the demand surge of 2020–2021. Each of those is a plausible cause of a hiring slowdown, and none of them is generative AI.

This does not prove that AI had no effect. It does mean that the cleanest natural experiment in the data — the level peak in the treatment month — is a coincidence of turning points, and that any paper reporting "information employment peaked the month ChatGPT launched" as evidence of an AI effect is reporting a level artefact.

6.2 The break-date model does not identify a date

A second check makes the same point from the other direction. If the trend-break method were identifying November 2022, the estimated post-slope should deteriorate as the assumed break date moves away from it. It does not. **Table 5** fits the same pre/post linear trend with five alternative break dates.

Table 5. Break-date sensitivity, information-sector index

Assumed break	Pre-slope (pts/yr)	Post-slope (pts/yr)
Nov 2019	+0.81	+0.16
Nov 2021	−0.06	−2.29
Nov 2022 (used)	**+0.86**	**−2.84**
Nov 2023	+1.12	−2.61
Nov 2024	+0.94	−2.84

Any break date from late 2021 onward produces a similarly negative post-slope, because the post-period is dominated by a single persistent decline. The method sizes the decline; it does not date it. November 2022 is used in this paper because it is the treatment date of interest, not because the data single it out. A reader who wants the date of the turn should use the growth series, which says August 2021.

6.3 Three shocks, one quarter

The information sector's decline begins in the same period as three distinct shocks:

Monetary policy. The Federal Reserve raised rates from March 2022 through July 2023. Long-duration assets, including the equity of technology firms, repriced. Firms whose valuation depends on distant cash flows cut discretionary spending first, and hiring is discretionary spending.

Pandemic normalisation. The sector index rose 12.8 per cent between January 2016 and November 2022, and a disproportionate part of that rise came after March 2020, when digital demand surged. Some of the subsequent decline returns employment to its pre-pandemic trajectory rather than departing from it. The counterfactual in Section 5.2 is an attempt to account for this, but it extrapolates a trend that includes the pandemic bulge, which biases the counterfactual upward and therefore the gap upward.

Generative AI. The third shock is the one of interest, and it is the only one that cannot be dated independently of the other two.

A fourth candidate is composition: the information sector is roughly 2 per cent of nonfarm payrolls, and its largest employers are a small number of firms whose hiring is visible and discretionary. A 12 per cent decline in a 2-per-cent sector is a 0.24-per-cent decline in aggregate employment — within the noise of a monthly payroll report.

6.4 Why health care is not a control

The difference-in-differences estimates in Table 4 put the information-sector decline at 25 percentage points relative to health care. That number is too large to be a treatment effect, and the reason is visible in the control.

Health care employment rose 25.3 per cent between January 2016 and September 2026, and it has not stopped rising. Its growth is driven by population ageing, insurance coverage, and a staffing shortage that predates generative AI by decades. Using it as a control assumes that, absent AI, information-sector employment would have grown at the same rate as health care employment — an assumption that is false on its face. The estimate is therefore an upper bound contaminated by the control's secular trend.

A better control would be a sector with similar cyclicality, similar rate sensitivity, and similar exposure but different AI adoption. The data set does not contain one. This is a limitation of sector-level analysis, not a fixable estimator choice.

6.5 What the exposure literature would need

Three things would move the question from exposure to adoption.

Establishment-level adoption data. Which firms bought which models, for which tasks, at what price, and what happened to their headcount in the following four quarters? The vacancy-posting evidence in Acemoglu et al. (2022) is the template; the generative-AI version of it does not yet exist in public form.

Hiring composition by seniority. A firm that expects AI to substitute for junior analysis stops hiring analysts before it fires them. Aggregate headcount would show nothing for several quarters; the entry-level share of new hires would show it immediately. This is the single most informative statistic available in principle and least reported in practice.

Within-occupation wage dispersion. If AI compresses the return to routine cognitive tasks, the wage distribution inside an occupation should widen before its mean moves. Occupation averages cannot see this; microdata can.

None of these require a better exposure index. They require different data.

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7. Five falsifiable predictions

The value of a framework is what it forbids. These are the observations that would change the conclusion of this paper, stated in advance so that the next four years of data can settle the question rather than extend the argument. Each prediction names the data, the metric, the threshold, and the observation that would refute it.

P1. Entry-level composition moves first. The share of new hires in AI-exposed occupations that are entry-level (less than two years of experience) falls by at least 3 percentage points relative to less-exposed occupations within eight quarters of measurable adoption, before any decline in total headcount in those occupations.

Data: linked employer–employee microdata or job-posting composition by seniority.

Refuted if: total headcount in exposed occupations falls without any change in the entry-level share, or the entry-level share is flat for eight quarters while adoption rises.

P2. Wage dispersion widens inside occupations. The 90/10 wage ratio within AI-exposed occupations rises by at least 0.05 log points relative to unexposed occupations over four years, with the lower tail falling in real terms.

Data: CPS ORG or administrative wage records.

Refuted if: the ratio is flat or narrows while exposure rises, or the effect appears only between occupations rather than within them.

P3. Adoption predicts hiring, not exposure. Establishment-level adoption intensity (spend on AI tools per employee) predicts a larger reduction in planned hiring than occupational exposure scores do, in a horse race where both are measured.

Data: firm software-spend data joined to establishment hiring.

Refuted if: exposure scores outperform measured adoption, which would imply the effect runs through expectations rather than use.

P4. Displacement is task-specific, not sectoral. Within a sector, employment declines are concentrated in occupations whose task bundles are most exposed, not spread across the sector.

Data: occupation × sector employment.

Refuted if: declines are uniform within sectors, which would indicate a sector-wide demand or financing shock rather than task substitution.

P5. The sector break narrows as rates normalise, unless AI is the cause. If the information-sector decline is rate-driven, it should stop or reverse within four quarters of the policy rate stabilising at a neutral level. If it is AI-driven, it should persist or deepen.

Data: information-sector employment, monthly; policy rate.

Refuted if: employment stabilises while adoption keeps rising — that would favour the rate explanation — or continues to fall after rates are stable, which would favour the AI explanation.

P5 is the closest thing to a decisive test available with public data, and it is the one this paper is least able to run: the rate cycle has not yet fully resolved.

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8. Threats to validity

Measurement. The analysis uses sector employment from the establishment survey, which counts jobs rather than workers and is subject to two rounds of revision. The most recent four months are provisional. Classification is by establishment industry, so a software engineer employed by a bank appears in finance, not information — which biases sector attribution in ways that are not observable in this data.

Index base. The 2019 = 100 base is a choice. A 2016 base would change the percentage declines; a 2021 base would make the post-period change smaller. The peak-to-trough percentage is robust to base choice because it is a ratio; the "since 2016" figure is not.

Treatment date. November 2022 is the public release date, not the adoption date. Enterprise adoption lagged release by a year or more, and much of it is not counted in any employment series. A later treatment date would make the pre-period include part of the effect; an earlier one would not.

Trend model. The counterfactual is a linear extension of a period that includes a pandemic. It is an accounting device of known crudeness. Its purpose is to size the break, not to estimate a counterfactual world.

Control validity. Health care is not a valid counterfactual for information employment, as Section 6.4 argues. No sector in this data is.

Ecological inference. Sector aggregates cannot speak to occupation, seniority, or task-level effects. The paper's claims are therefore about sectors and aggregates; any extension to occupations is unsupported.

Publication and confirmation risk. The choice to write a paper about a visible break invites the reading that the break is the finding. The paper's finding is the opposite: the break is real, and it does not identify the cause.

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9. Reproducibility

The repository contains one data file, one analysis script, and the figures it generates.

```
data/labor-model.csv        # 129 rows, 25 columns, BLS series
analysis/analyze.py         # stdlib only; no network; deterministic
analysis/stats.json         # every number quoted in this paper
figures/fig1..fig5.svg      # regenerated by the script
```

Run:

```
python3 analysis/analyze.py
```

The script prints the summary statistics and rewrites `analysis/stats.json`; the figures are rewritten in place. Every number in Sections 5–7 traces to a field in `stats.json`. The data file is the current BLS vintage as of November 2026; a later vintage will produce slightly different recent-month values and is the correct behaviour.

Two checks a reader can run without reading the code: confirm that `stats.json.trend_break.information.gap_pct` equals −14.3, and that `stats.json.sectors.information.peak.date` equals 2022-11-01. Those two numbers carry most of the paper's argument.

---

10. Conclusion

The generative-AI period has produced a large, visible, persistent decline in information-sector employment in the United States. The sector peaked in November 2022, 12.1 per cent below its current level, and the break is 15.9 index points — 14.3 per cent — larger than its pre-2022 trend would have predicted. This is the largest sectoral break in the data since the pandemic.

The same data show that the break cannot be attributed to generative AI. The sector's growth rate began falling in August 2021, fifteen months before ChatGPT was released. The break-date model cannot distinguish November 2022 from any later date. The other shock that begins in that quarter — the most aggressive tightening cycle in four decades — operated on exactly the firms that make up the sector. And the sector that should show the largest effect if task exposure were the mechanism, professional and business services, shows a smaller one than the sector that produces the technology.

None of that is evidence that AI will not displace labour. It is evidence that four years of sector aggregates cannot answer the question, and that the exposure indices now circulating in policy writing are being asked to answer a question they were never built for. An exposure score measures what a model can do; employment data measures what firms did. Between them sits the adoption decision, and it is the adoption decision that policy should be measuring.

The five predictions in Section 7 are the constructive part. Each names a statistic that would move first if the displacement thesis were correct, and each has a stated refutation. If those statistics move as predicted over the next four years, the exposure literature will have earned its policy influence. If they do not, the literature will need to explain why the capability did not become a decision.

Appendix A. Data dictionary

Column	Definition
`date`	Month, first of month, ISO format
`unrate`	Unemployment rate, per cent, SA
`unemployed_k`	Unemployed persons, thousands, SA
`openings_k`	Job openings, thousands, SA (JOLTS, one-month lag)
`v_u_ratio`	Openings per unemployed person
`payrolls_k`	Total nonfarm payrolls, thousands
`payrolls_idx`	Payroll index
`payroll_yoy`	Payrolls, 12-month per cent change
`wages_nom`	Average hourly earnings, total private, USD

Column	Definition
`cpi`	CPI-U, all items, NSA
`wage_yoy_nom`	Nominal wage growth, 12-month per cent
`cpi_yoy`	CPI inflation, 12-month per cent
`wage_yoy_real`	Real wage growth, 12-month per cent, (1+nom)/(1+cpi)−1
`idx_info`	Information employment index, 2019 = 100
`idx_profbiz`	Professional and business services index, 2019 = 100
`idx_mfg`	Manufacturing index, 2019 = 100
`idx_health`	Health care and social assistance index, 2019 = 100
`yoy_info`	Information employment, 12-month per cent change
`yoy_profbiz`	Professional and business services, 12-month per cent
`yoy_mfg`	Manufacturing, 12-month per cent
`yoy_health`	Health care and social assistance, 12-month per cent
`latest_flag`	1 on the final row
`latest_valid_flag`	1 on the last row carrying JOLTS data

Appendix B. Figures

Figure 1. Employment by sector, index 2019 = 100, January 2016 – September 2026. Vertical line: November 2022.

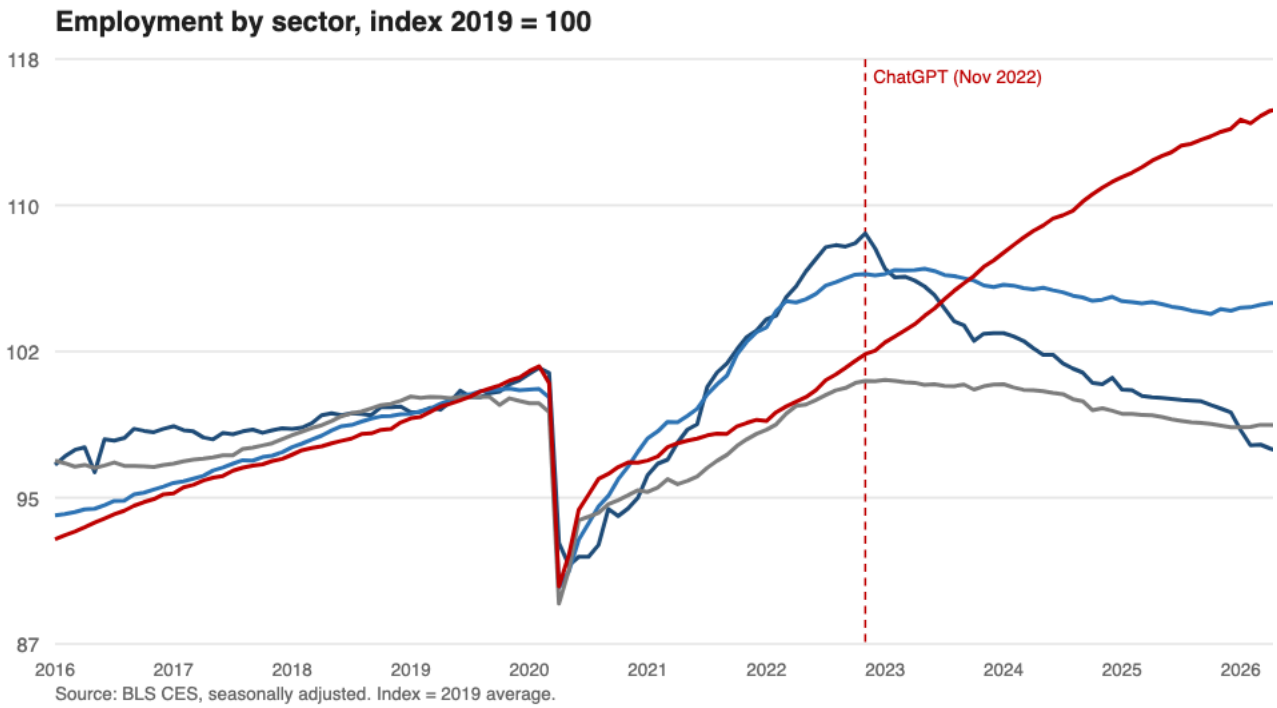


Figure 2. Information-sector employment and pre-trend counterfactual. The counterfactual extends the January 2016 – October 2022 linear trend. It is an accounting device, not a forecast.

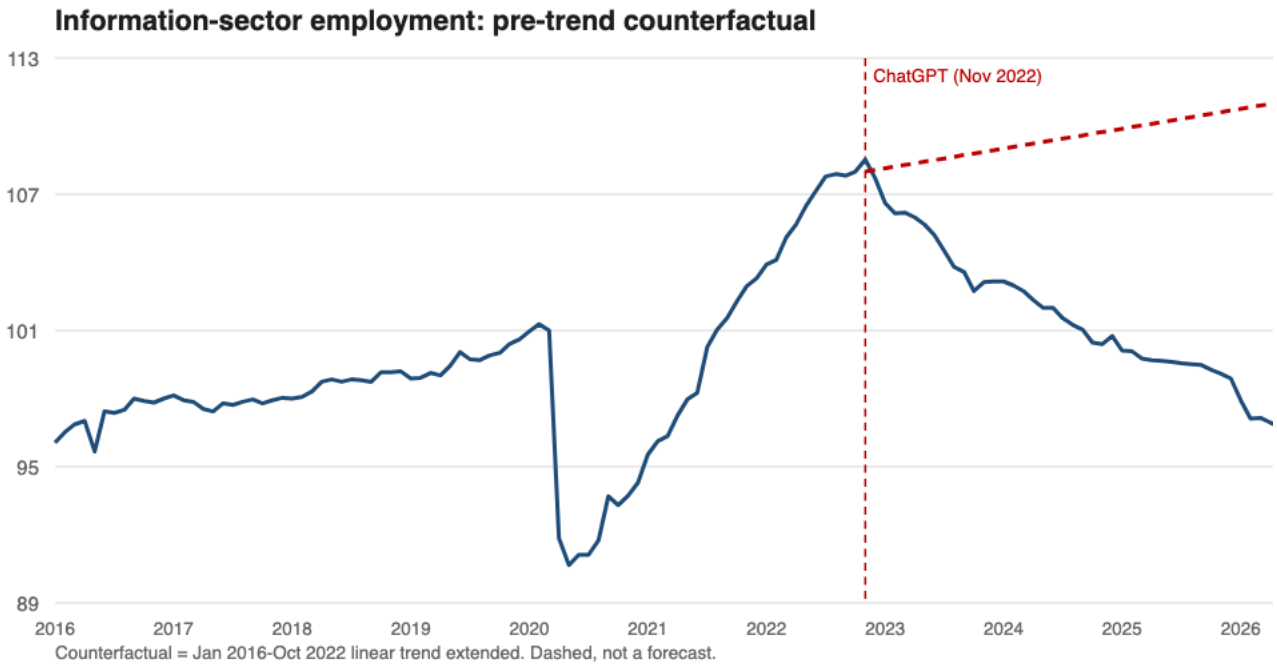


Figure 3. Twelve-month employment growth by sector. Shaded: post-November 2022.

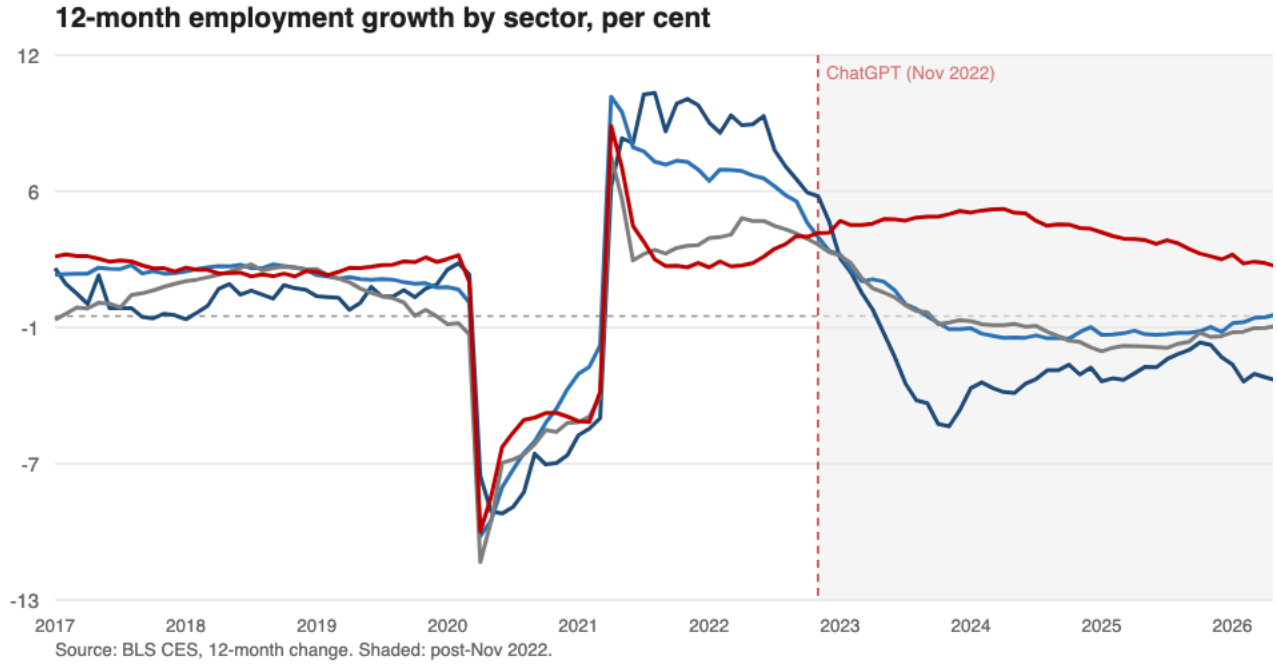
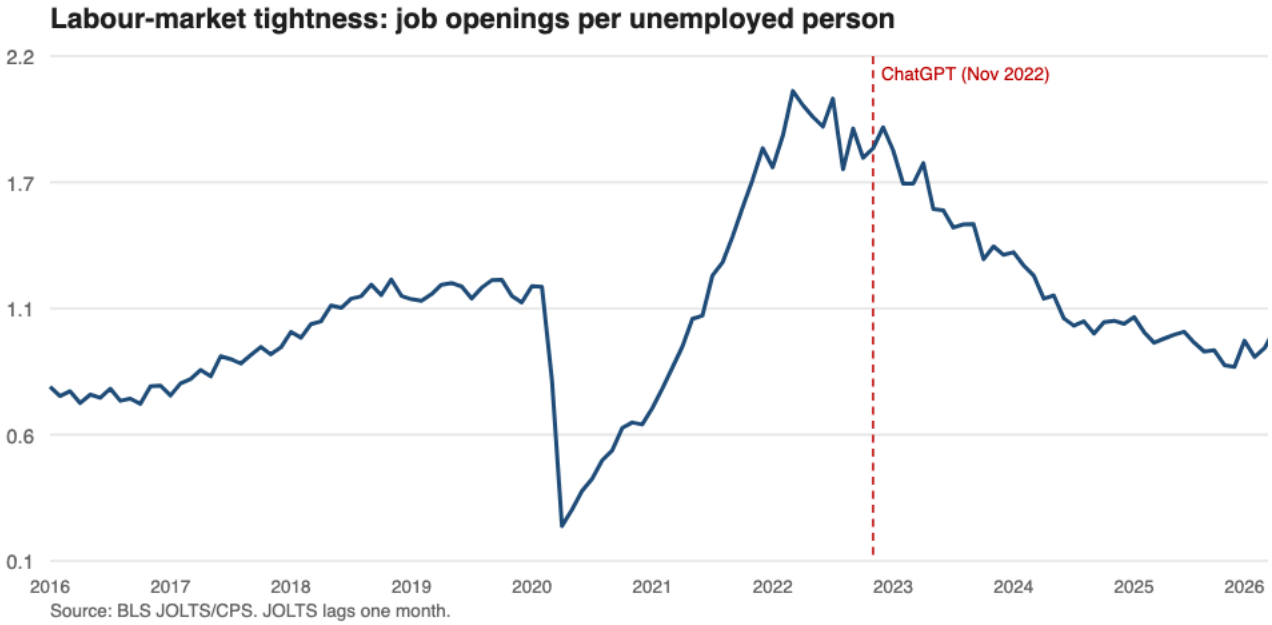


Figure 4. Nominal and real wage growth, 12-month per cent change.



Figure 5. Job openings per unemployed person, January 2016 – August 2026.



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